

SkillConnect

Next-Gen Peer-to-Peer Skill Exchange & Mentorship Platform

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Repository: github.com/Yashvardhan4646/skill-o-connect

Vision: Democratizing 1-on-1 Mentorship & Collaborative Learning

Problem Statement: Gaps in Skill Acquisition

■ ■ Exorbitant Costs

1. High Cost of Mentorship: Professional tutoring and bootcamp platforms charge exorbitant fees, pricing out students and everyday learners.

■ Passive Tutorials

3. Passive Tutorial Fatigue: Standard video courses lack hands-on accountability and collaborative buildathon projects.

■ Geographic Isolation

2. Geographic Isolation: Learners struggle to find nearby peers or local mentors in their city who share complementary skill goals.

■ Discovery Deficit

4. Discovery Deficit: Discovering compatible mentors based on skill synergy, location, and proven peer ratings is difficult without smart algorithms.

Our Solution: Zero-Cost Skill Barter Ecosystem

■ Reciprocal Skill Barter

Reciprocal Skill Barter: Exchange your expertise (e.g. React.js) to learn another skill (e.g. Graphic Design) with zero monetary fees.

■ Algorithmic Matchmaking

Jaccard Algorithmic Matchmaker: Smart algorithm calculating skill overlap, category tags, and optional GPS distance to surface optimal partners.

■ Role Differentiation

Mentors vs. Mentees (Scholars): Role differentiation allowing mentees to display learning wishlists and mentors to gain leaderboard rank.

■ Direct DM Messaging

Direct Realtime Messaging: Real-time Supabase messaging system for seamless 1-on-1 communication and session coordination.

Key Platform Features & Architecture

■ Collaborative Buildathons

■ **Community Challenges Board:** Team buildathon board where mentors post micro-projects for hands-on collaboration.

■■ Proximity GPS Mapping

■ ■ **Interactive Proximity Map:** Leaflet OpenStreetMap integration rendering local mentor pins and distance metrics.

■ URL Project Embeds

■ **Rich Embed Project Showcase:** Transform GitHub, Figma, and CodePen links into embedded preview cards (zero upload hazards).

■ Reputation & Leaderboard

■ **Verified Achievement Badges:** Reputation system awarding Master Sensei, Founder VIP, 5-Star, and Rapid Exchanger badges.

Technical Stack & Infrastructure

Layer	Technology Stack	Core Capability
Frontend UI/UX	HTML5, Vanilla CSS3 (Custom Tokens), ES6 JavaScript	Responsive SPA, HSL dark/light modes, micro-animations
Backend & DB	Supabase PostgreSQL 15, Auth, Realtime WebSockets	Idempotent schemas, RLS security policies, instant DM sync
Mapping Engine	Leaflet GIS API & OpenStreetMap	Dynamic pin placement & city geolocation search

Algorithmic Innovation: Smart Match Engine

1. **Jaccard Skill Similarity Index:** Calculates intersection over union for offered and requested skill sets to quantify match compatibility.
2. **GPS Proximity Distance Formula:** Haversine formula calculation for distance filtering between local mentor/mentee pins.
3. **Leaderboard Qualification Filter:** Filters out mentee-only profiles from mentor standings to preserve podium integrity.
4. **Content & Role Safety Rules:** Role-based access control and strict data validation across profiles and proposals.

Product Roadmap & Future Growth

Phase 1: NYC CodeQuest MVP

Phase 1: NYC CodeQuest MVP: Core skill exchange marketplace, Supabase DM inbox, live map, and challenge board.

Phase 2: Campus Expansion

Phase 2: Campus Expansion: University chapter onboarding, developer club integrations, and automated badge issuance.

Phase 3: AI Ecosystem

Phase 3: AI Ecosystem: LLM-powered personalized curriculum generation and automated mentor-mentee schedule optimizer.

Conclusion & Contact

SkillConnect bridges the gap between eager learners and passionate mentors worldwide. By combining reciprocal skill barter, direct messaging, smart matchmaking algorithms, and collaborative buildathons, we democratize quality 1-on-1 education.

Project Links & Contact:

- **Author:** Yashvardhan
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- **GitHub Repository:** <https://github.com/Yashvardhan4646/skill-o-connect>
- **Event:** NYC CodeQuest 2026

Thank you for your time! We welcome questions and feedback.